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The Oxygen Dependence of Mitochondrial Oxidative Phosphorylation Measured by a New Optical Method for Measuring Oxygen Concentration*

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Oxygen-dependent quenching of phosphorescence has been used to measure the dependence of mitochondrial oxidative phosphorylation on oxygen concentration in suspensions of isolated rat liver mitochondria. An instrument has been designed which simultaneously monitors the phosphorescence lifetime of a fluorophor and the reduction of cytochrome *c* by dual wavelength spectrophotometry. The phosphorescence lifetime method gives very rapid (<100 ms) measure of the oxygen concentration (Vanderkooi, J. M., Maniara, G., Green, T. J., and Wilson, D. F. (1987) *J. Biol. Chem.* 262, 5476–5482) from concentrations characteristic of air-saturated media to as low as 2×10^{-8} M. The results may be summarized as follows.

For well coupled rat liver mitochondria at pH 7.0 and in the presence of ATP, as the oxygen concentration was lowered, increased cytochrome *c* reduction was observed to begin at oxygen concentrations greater than 20 μM.

For mitochondria in the presence of uncoupler, cytochrome *c* reduction began at oxygen concentrations less than 1.0 μM.

The oxygen dependence of reduction of cytochrome *c* in well coupled mitochondria treated with ATP was strongly dependent on the pH of the suspending medium. Reduction of cytochrome *c* began at higher oxygen concentrations as the pH was made more alkaline.

The oxygen concentration for half-maximal respiratory rates was much larger for well coupled mitochondria treated with ATP (approximately 0.7 μM) than for mitochondria treated with uncoupler (less than 0.1 μM).

It is concluded that the oxygen dependence of mitochondrial oxidative phosphorylation is such that mitochondria could function in their proposed role of tissue oxygen sensors for regulation of such diverse functions as local blood flow and electrical activity in the carotid body.

Mitochondrial oxidative phosphorylation is responsible for the metabolism of most of the oxygen utilized by higher organisms. This oxygen is required for production of the adenosine triphosphate (ATP) used to provide energy for biosynthesis, transport, and mechanical work. Clearly an understanding of the effects of oxygen deficiency includes a thorough knowledge of the dependence of mitochondrial oxidative phosphorylation on the concentration of oxygen. The lack of methods for reliable and rapid measurement of oxygen concentrations in biological systems has limited our ability to obtain these data. In general, each of the previously available methods utilized has been effective for only part of the range of oxygen concentrations of interest. Oxygen electrodes, for example, are very useful for oxygen concentrations from those of air-saturated media down to about 5 μM. Below this concentration accuracy has been obtained (see for example Ref. 1), but usually only with modifications which make the response time too slow for many experimental purposes.

We have recently developed a method for measuring oxygen by its ability to quench the phosphorescence of selected luminophores (2–4). This method is capable of accurately measuring oxygen concentrations from more more than 10^{-3} M to less than 10^{-8} M. The phosphorescence probe molecules are soluble in biological media and respond to changes in oxygen concentration in microseconds. Only a few milliseconds are required to take the data for each oxygen measurement, and the measurements can be repeated at least twice per second. Since this method can continuously monitor oxygen concentration through the range of physiological interest, we have reexamined the oxygen dependence of mitochondrial oxidative phosphorylation using suspensions of isolated mitochondria.

The oxygen concentration in the suspending medium and reduction of cytochrome *c* were monitored simultaneously as functions of oxygen concentration. The data show that mitochondrial oxidative phosphorylation is dependent on oxygen concentrations up to at least 20 μM at pH 7.0 and the oxygen dependence becomes markedly greater as the pH of the suspending medium is made more alkaline. The oxygen concentration for half-maximal respiratory rate at pH 7.4 is approximately 0.7 μM for well coupled mitochondria and decreases to less than 0.1 μM when the mitochondria are uncoupled.

MATERIALS AND METHODS

Oxygen concentration was measured by its quenching of the phosphorescence of selected luminophores (2–4). A xenon flash lamp with a flash duration (width at half-height) of less than 2 μs was used to excite palladium coproporphyrin, the luminophore chosen for these studies, to its excited triplet state and to measure the phosphorescence lifetime by the decay of light emission following the flash (see Ref. 4 for more technical detail). The measured phosphorescence lifetime was used to calculate oxygen concentration from the Stern-Volmer relationship:

$$T_0/T = 1 + T_0 k_q[O_2] \quad (1)$$

where T_0 is the lifetime when the oxygen concentration ($[O_2]$) was zero, and k_q is the collisional quenching coefficient. The Stern-Volmer

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relationship is accurately obeyed for oxygen quenching of the phosphorescence of palladium-coproporphyrin (4).

The sample chamber was designed to hold a 1-cm square glass cuvette mounted on a magnetic stirrer. The sample in the cuvette was stirred continuously using a small (0.8 × 0.15 cm) magnetic stirring bar, and the top was sealed with a ground glass stopper having a small (0.8 mm in diameter and 1.2 cm high) exit hole in the top. All air bubbles were removed through this hole, and the liquid sample filled it to the top. This gave an unstirred liquid column of 1.2 cm to prevent oxygen leakage into the sample chamber while permitting reagents to be introduced with a Hamilton syringe.

The optical components were arranged as shown in Fig. 1. Both oxygen concentration and cytochrome *c* reduction were measured in the same volume of sample. This, however, required that the measurements be made using a time-sharing arrangement. An air turbine spectrophotometer designed and constructed in the shops of the Johnson Research Foundation was used for measurements of cytochrome *c* reduction (see Fig. 2). The air-driven turbine had four positions for mounting interference filters. Filters with maximal transmission at 540 and 550 nm and transmission bandwidths at half-height of 1.5 nm were mounted in *positions 1* and *2* of the turbine, while *positions 3* and *4* were blank. A stream of air rotated this turbine at 60 Hz, resulting in each filter position passing between the light

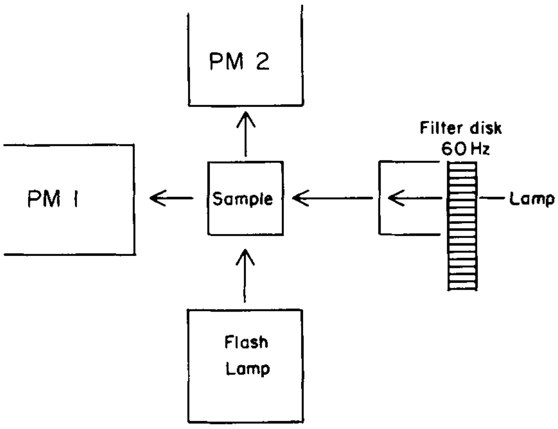

FIG. 1. **A schematic diagram of the optical arrangement of the instrument for simultaneous measurement of oxygen concentration and cytochrome *c* reduction.** The sample chamber holds a square glass cuvette with a 1-cm light path. The light from the light source for the dual wavelength spectrophotometer passes through an air-driven turbine with interference filters to select the wavelengths of the measuring light, passes through the sample, and falls on a photomultiplier (*PM 1*). Light from the flash lamp for phosphorescence lifetime measurements passes through optical filters which pass light from 380 to 440 nm with maximal transmission at 400 nm. Light emitted from the sample is filtered to pass only light of greater than 650 nm and falls on a red sensitive photomultiplier (*PM 2*).

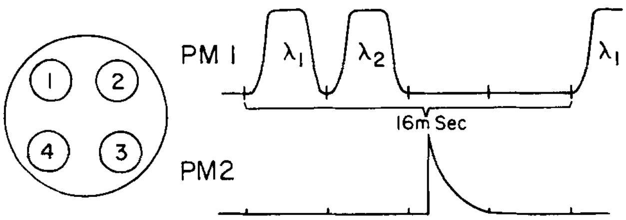

FIG. 2. **Time sharing by the absorption and phosphorescence measurements.** The geometry of the interference filters mounted in the air turbine is shown on the left. Filters passing light at 540 and 550 nm were mounted at *positions 1* and *2* while *positions 3* and *4* were blank. The photocurrent for the two photomultipliers are schematically shown for one revolution of the turbine (the rotational rate was adjustable but for most experiments it was approximately 60 Hz or 17 ms/revolution). The flash lamp was fired at the time coinciding with *position 3* by a signal generated by the turbine position.

source and the sample 60 times/s. The photocurrent from PM 1[1] was demodulated, the signal for 540 nm subtracted from that for 550 nm, and the difference amplified. The resulting signal was a measure of the changes in light transmission at 550 nm relative to that at 540 nm. A position indicator signal from the air turbine was used to fire the flash lamp at position 3. Optical filters were used to limit the excitation wavelengths from the flash lamp to between 380 and 440 nm. The light reaching PM 2 was filtered such that no signal from the light at 550 and 540 nm or the flash lamp *per se* was observed. The response of PM 2 was due to fluorescence and phosphorescence arising from the sample and having a wavelength greater than 650 nm. The signal from PM 2 was digitized by a 20-MHz, 8-bit PCTR-160 A/D card from General Research Corp., McLean, VA, and discharge of the flash lamp was detected by the rising edge of the photomultiplier output (threshold detection). The data were transferred to the microcomputer by direct memory access, and the software allowed the output from any number of flashes to be averaged prior to calculation of the phosphorescence lifetime.

The measured absorbance changes at 550 nm minus 540 nm were made during the same time interval as the oxygen measurements. A demodulator circuit separated the signals for the two wavelenghts, subtracted the two, and amplified the difference for output of ± 50 mV. This signal was digitized using an analogue interface card (DM-100PC, Dataq Instruments, Inc. Akron, OH), also resident in the microcomputer, during the time that the phosphorescence data was being taken. Both the phosphorescence decay and absorbance changes at 550 nm minus 540 nm were taken over 5 or 10 cycles (five flashes occurred in approximately 80 ms). Each data set was averaged, and the values of T, cytochrome *c* reduction, and time were stored in arrays in the computer memory. The points were also plotted on the monitor to permit real time monitoring of the progress of the experiment. Using a 12-MHz PC'S Limited 286[12] microcomputer and a program written in C language, one set of values could be taken each 0.6 s.

Data analysis and graphics were carried out using a software program, Asystant, from Macmillian Software Co., New York.

Isolation of Mitochondria from Rat Liver—The mitochondria were isolated essentially by the method of Schneider (5) except that the isolation medium was 0.225 M mannitol, 0.075 M sucrose, and 1 mM EGTA with a final pH of 7.4. The isolated mitochondria were maintained on ice as a suspension, of approximately 60 mg of protein/ml, in the isolation medium. In order to measure oxidative phosphorylation, the mitochondria were diluted to the protein concentration indicated in the figure legends. The assay medium was similar to the isolation medium but contained 10 mM 4-morpholinopropanesulfonic acid adjusted to the indicated pH using Trizma base. Bovine serum albumin was used in the suspending medium at 0.4% by weight for two reasons: 1) When palladium coproporphyrin is bound to bovine serum albumin, the possible interactions with the mitochondria are minimized. Indeed, addition of mitochondria had no effect on the phosphorescence lifetime or the oxygen-quenching constant. Conversely, addition of the probe had no effect on the mitochondrial respiratory rate, respiratory control, or the ADP/O ratio. 2) Self-quenching was minimized and differences in the concentration of added probe (in the low concentration range required for these experiments) did not affect either T_0 or k_q.

Two different types of substrate were used. A combination of succinate (8 mM), glutamate (8 mM), and malate (1 mM) was used to simulate oxidation of the citric acid cycle intermediates. Ascorbate (8 mM) and *N,N,N',N'*-tetramethyl-*p*-phenylenediamine (TMPD) were used to examine the respiratory behavior when cytochrome *c* was reduced directly by a nonenzymatic electron donor. In the latter case, oxidized cytochrome *c* (*c*) is reduced in a simple second order reaction:

$$\text{TMPD} + c^{3+} \rightarrow c^{2+} + \text{Wurster's blue} \quad (2)$$

for which the rate expression is:

$$v = k[\text{TMPD}][c^{3+}] \quad (3)$$

The Wurster's blue which is formed is rapidly reduced to TMPD by ascorbate, and its steady state concentration remains too low to interfere with the optical measurements. The second order rate constant, k, has a measured value of 1.38×10^5 M^{-1} s^{-1} (6).

The data presented in the figures are representative data taken

[1] The abbreviations used are: PM, photomultiplier; EGTA, [ethylenebis(oxyethylenenitrilo)]tetraacetic acid; TMPD, *N,N,N',N'*-tetramethyl-*p*-phenylenediamine.

from more than 30 different mitochondrial preparations over a 6-month period. Representative data mean that similar data were obtained in every mitochondrial preparation tested, which was more than 10 for the experiments described in each figure.

RESULTS

Measurements of Oxygen Consumption by Suspensions of Isolated Mitochondria—Rat liver mitochondria were suspended in a mannitol/sucrose assay medium at pH 7.4 and the respiratory rate measured. Phosphorescence lifetimes were measured and were plotted on the microcomputer display. These values of T, T_0, and k_q were used to calculate the oxygen concentrations. The data for a typical experiment, as they appeared on the computer display, are shown in Fig. 3. Ascorbate was present in the suspending medium, and 75 μM TMPD was added to initiate mitochondrial respiration. Each measured value of T was calculated from the decay curve obtained by averaging the phosphorescence decay curves for 10 flashes (170 ms). The phosphorescence lifetime was 37 μs at 250 μM O_2 and increased to 440 μs at zero oxygen. In the experiment shown in Fig. 3, the oxygen concentration was approximately 270 μM at the start of the measurements. After anaerobiosis, the medium was reoxygenated by adding 200 μM H_2O_2 (which was decomposed by catalase to form O_2). Two cycles of oxygen depletion are shown. During the experiment a total of 880 measurements were made in approximately 600 s. The data points are plotted on the monitor by sample number and not time. The software results in some variation in the time required to calculate the phosphorescence lifetime and thereby some variability in the decrease in oxygen concentration from sample to sample. Clock readings were made at the same time as the data for each phosphorescence lifetime and were stored in a separate file. Time dependences were obtained using the clock readings. The cycles of oxygen depletion are highly reproducible.

Simultaneous Measurement of Oxygen Concentration and Cytochrome c Reduction—The dependence of cytochrome *c* reduction on oxygen concentration was measured in suspensions of mitochondria at pH 7.4, oxidizing either succinate plus glutamate or ascorbate plus TMPD. An example of

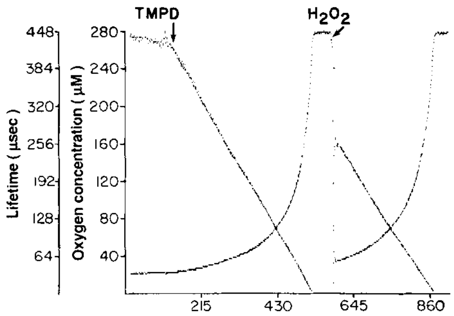

FIG. 3. **Measurement of oxygen concentration by phosphorescence lifetime in suspensions of isolated rat liver mitochondria.** Rat liver mitochondria were suspended at approximately 1 mg of protein/ml in assay medium of pH 7.4. The medium that contained 8 mM ascorbate·TMPD (75 μM) was added to initiate respiration. The measured values of phosphorescence lifetime are shown as they appear in real time during the experiment. At the end of the experiment the oxygen concentrations were calculated as described under "Materials and Methods" and also plotted on the screen. Each point is an individual measurement of phosphorescence lifetime. A total of 880 points was taken in 600 s. The figure shows the data as it appeared on the monitor and this means the values are plotted against sample number. The rate at which the samples were taken was not constant, but the real time is read from a clock and stored by the computer for later analysis.

experimental data for mitochondria at pH 7.4 and in the presence of ATP is shown in Fig. 4*A*. As the oxygen concentration decreased below approximately 50 μM, there was progressive reduction of cytochrome *c*. Reoxidation of the suspension at any point resulted in reoxygenation of cytochrome *c* to the value for the higher oxygen concentration. Most preparations of mitochondria were able to undergo three to four reoxygenation cycles with consistent results, as long as they were not subjected to extended periods of anaerobiosis (>1–2 min). In a few preparations there was a progressive increase in the respiratory rate with the time of incubation. When this occurred the data were not analyzed. Although preparations of isolated rat liver mitochondria have catalytic activity, for experiments in which relatively low protein concentrations (less than about 1 mg/ml) were used, exogenous catalase was added to ensure decomposition of the H_2O_2 in less than 2 s.

The relationship between cytochrome *c* reduction and oxygen concentration is shown in Fig. 4*B*. The changes in absorbance at 550 nm minus 540 nm were used to calculate reduction of cytochrome *c* as percent of the total present in the sample. The percent reduction of cytochrome *c* was then plotted directly against oxygen concentration. With succinate plus glutamate as substrate, cytochrome *c* was approximately 18% reduced at high oxygen concentrations and became noticeably more reduced by the time the oxygen concentration decreased to 30 μM. Similar data were obtained when the substrate was ascorbate plus TMPD.

The Effect of pH of the Suspending Medium on the Oxygen Dependence of Cytochrome c Reduction—When measurements of cytochrome *c* reduction were made in the presence of ATP, for similar respiratory rates, reduction of cytochrome *c* was much greater at pH values near 8 than near 7 (7, 8, 10). This is an important expression of the oxygen dependence of mitochondrial oxidative phosphorylation which could be readily tested using this method. Experiments were therefore carried out in which the mitochondria were suspended in media of different pH values and both oxygen concentration and cytochrome *c* reduction measured. Data from experiments at pH 7.0 and 7.8 are shown in Fig. 5, *A* and *B*, and can be compared with the data for pH 7.4 (Fig. 4*B*). Cytochrome *c* reduction began at much higher concentrations of oxygen at pH 7.8 than at pH 7.0. Although the difference cannot be precisely quantitated from these experiments, increases of similar magnitude in cytochrome reduction occurred at oxygen concentrations of 20, 35, and >100 μM for pH values of 7.0, 7.4, and 7.8, respectively.

The Effect of Metabolic Energy State on the Oxygen Dependence of Cytochrome c Reduction and Respiration—Treatment of mitochondria with uncoupler caused a large decrease in the oxygen concentrations at which increased reduction of cytochrome *c* and decreased respiratory rates occurred. This is most easily observed in the reduction of cytochrome *c*. In well coupled mitochondria at pH 7.4, reduction began by the time the oxygen concentration had fallen below 30 μM (Fig. 4), whereas in the presence of uncouplers reduction of cytochrome *c* did not begin until the concentration of oxygen had fallen below about 1 μM (Fig. 6). The value for uncoupled mitochondria is approximate since it was limited by the analogue circuit used for cytochrome *c* measurements. The signal to noise ratio for the cytochrome *c* measurements required the use of a response time (0–95%) of 2 s. Lowering the respiratory rate by diluting the mitochondria was not practical, because that decreased the absorbance change and required a correspondingly longer time constant.

The dependence of the respiratory rate of well coupled

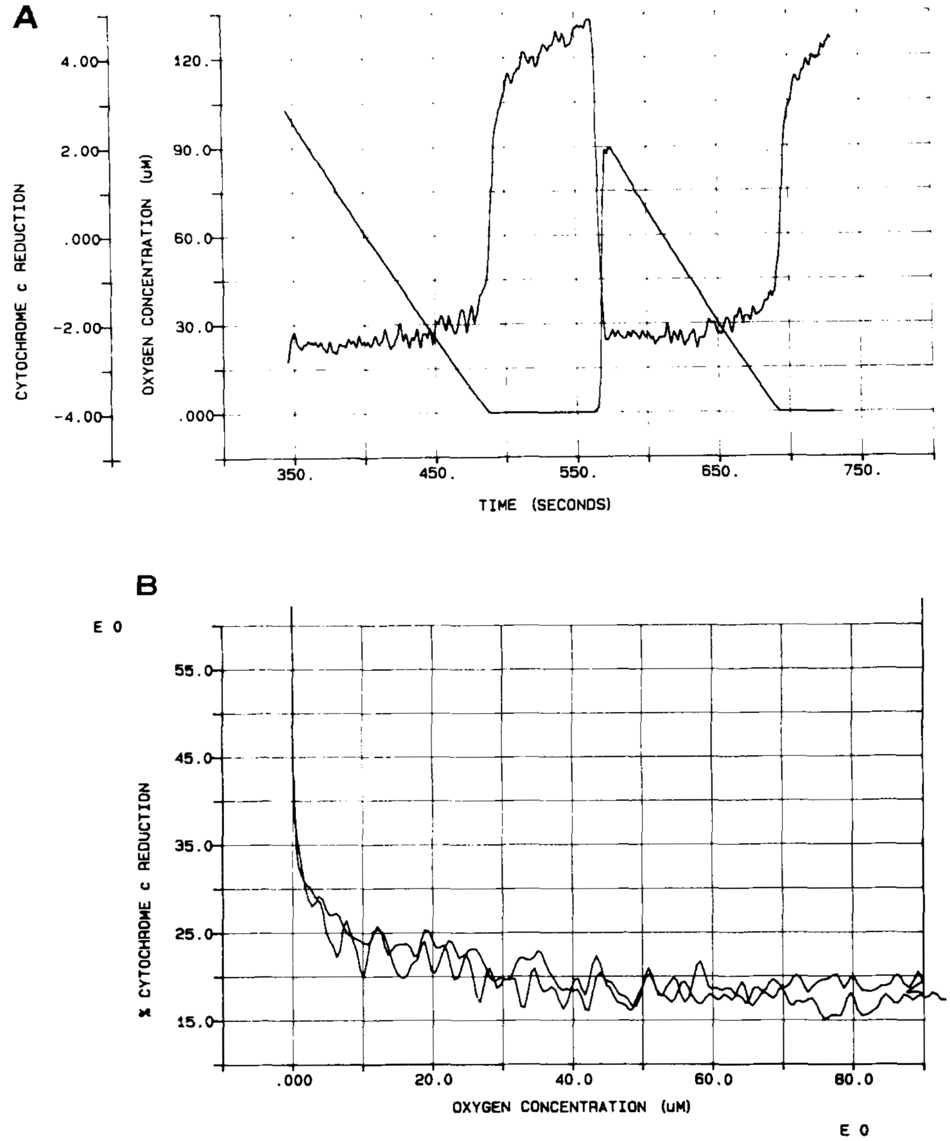

FIG. 4. **Simultaneous measurement of oxygen concentration and cytochrome *c* reduction in suspensions of mitochondria.** Mitochondria were suspended as described in the legend of Fig. 3 in a medium containing 0.87 mM ATP, 8 mM succinate, 8 mM glutamate, and 1 mM malate. The data for oxygen concentrations less than 100 μM are shown. *A*, the data for two cycles of oxygen depletion are shown. After anaerobiosis was attained in the first cycle, an aliquot of H_2O_2 was added to give an oxygen concentration of approximately 90 μM after catalytic decomposition of the H_2O_2 to O_2. *B*, a replot of the data as percent reduction of cytochrome *c* against the concentration of oxygen.

mitochondria oxidizing succinate plus glutamate on oxygen concentration was readily observable at oxygen concentrations below 5 μM (Fig. 7, *A* and *B*). Measurements of the oxygen concentration in a suspension of mitochondria in the presence of ATP as a function of time are shown in Fig. 7*A*, with emphasis on the measurements below approximately 16 μM. Fig. 7*B* shows the rate of respiration as a function of oxygen concentration and the fit of the data to the relationship:

$$v = V_m \times [O_2]/(P_{50} + [O_2]) \quad (4)$$

where v is the rate of respiration (dO_2/dt), V_m is the respiratory rate with saturating oxygen concentration, and P_{50} is the oxygen concentration at half-maximal respiratory rate. In analyzing data from six different mitochondrial preparations at pH 7.2–7.4, the P_{50} values ranged from 0.5 to 0.9 μM. The P_{50} value decreased when the mitochondria were treated with uncoupler (Fig. 8). Due to the stimulation of respiration which occurred upon addition of uncoupler (8–10-fold), special experimental protocols were used to set upper limits on this value. The phosphorescence lifetime data for each point were taken in 80 ms, but data processing required approximately 0.4 s. This made it possible to measure the phosphorescence lifetime only twice per second. In order to have sufficient data points to resolve a P_{50} of 0.1 μM, it was necessary to dilute the mitochondria until the maximal respiratory rate was 0.1 μM/s or less. In addition, the media were bubbled with nitrogen prior to addition of mitochondria in order to lower the initial oxygen concentrations to below 40 μM and to decrease the time required for these low respiratory rates to deplete the oxygen. As may be seen from Fig. 8, the rate of oxygen consumption by uncoupled mitochondria was independent of oxygen concentration to less than 0.1 μM. Analysis indicated that the P_{50} value was certainly below 0.1 μM and probably less than 0.05 μM. This was, however, as low as could be resolved in this experiment because the probe sensitivity was limited to $\pm$0.02 μM oxygen.

DISCUSSION

Comparison of the Oxygen Dependence of Mitochondrial Oxidative Phosphorylation with the Results in the Literature Using Other Methods—An examination of the literature indicates that the data fall into two groups which are fundamentally different. We will focus on two differences critical to understanding the physiological role of mitochondria: 1) The P_{50} for well coupled mitochondria respiring at a high [ATP]/[ADP][P_i] ratio. One group reports a value of less than 0.05 μM (13–15) while the other gives a value of 0.5–0.9 μM (8–12, this paper). 2) The effect of mitochondrial metabolic state on the P_{50} for respiration and cytochrome reduction.

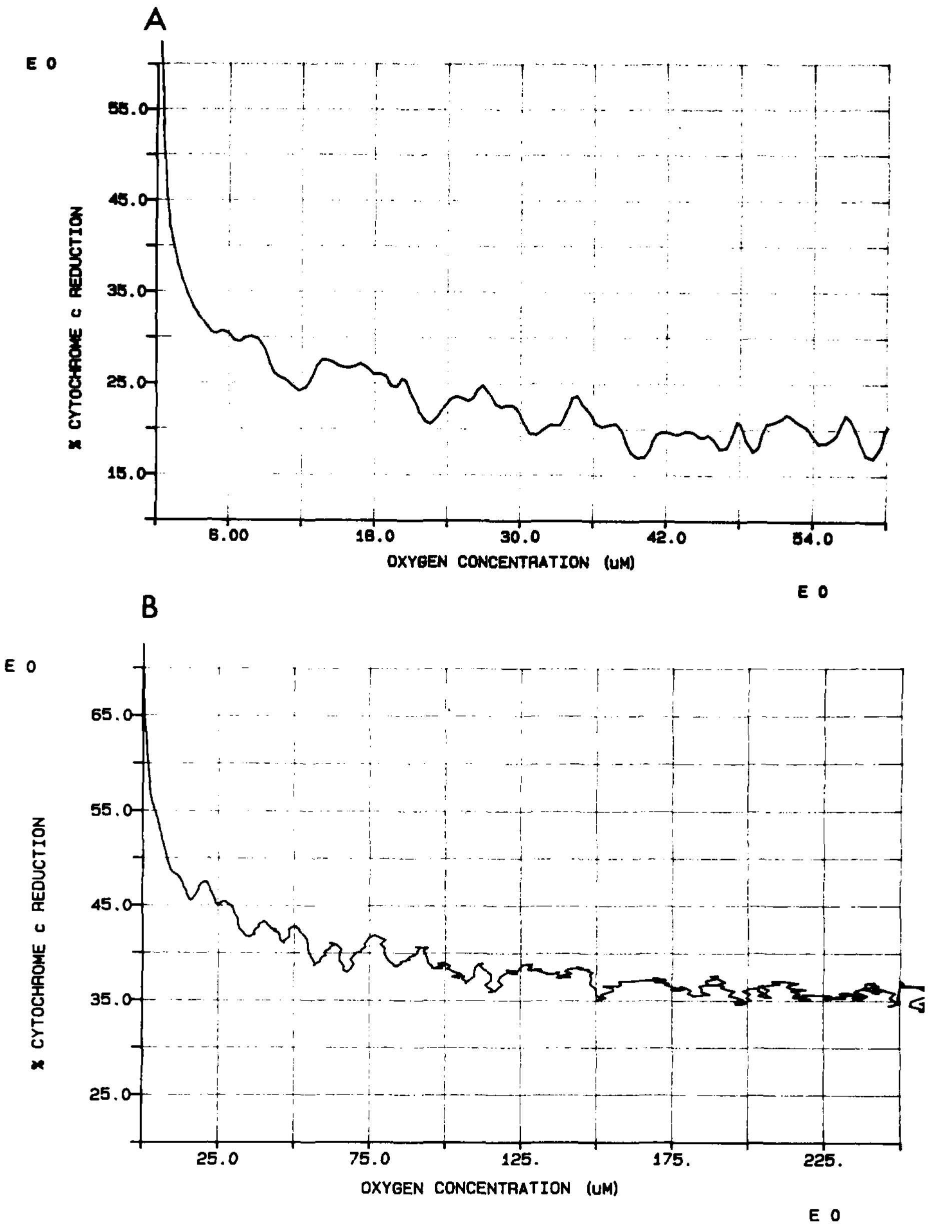

FIG. 5. **The effect of pH on the oxygen dependence of cytochrome *c* reduction.** The experimental conditions were the same as for Fig. 4 except that the pH of the assay medium was 7.0 (*A*) or 7.8 (*B*).

One group reports the P_{50} increases by severalfold (13–15) and the other that it decreases by more than 10-fold (8–12, this paper) when mitochondria are uncoupled (compared to the value at high [ATP]/[ADP][P_i]). We will briefly discuss the experimental methods used in an attempt to identify possible reasons for this large discrepancy in experimental results.

The experimental conditions used in this work were well suited to the study of mitochondrial respiration. The phosphor (palladium coproporphyrin) used as an oxygen indicator was dissolved in the medium in which the mitochondria were suspended. Under these conditions the phosphorescence is quenched by oxygen with a rate constant of 1×10^8 M^{-1} s^{-1}. This means the limiting response time for measuring changes in oxygen concentration is 10 ms at 10^{-6} M oxygen. In the present experiments, the values of T are calculated from the data for either 5 or 10 flashes (five flashes at 60 Hz occur in less than 90 ms), five flashes giving a time resolution of 0.1 s. The T_0 value of 1.2 ms combined with the k_q above, allows oxygen concentrations from above 250 μM to about 10^{-8} M to be measured. The results at high oxygen concentrations were similar to previous measurements from this laboratory using oxygen electrodes (see, for examples, 7–10). The present measurements, however, extend to much lower oxygen concentrations (0.02 μM *versus* at least 2 μM), and only in the present experiments could the P_{50} for respiration be resolved. The data obtained are unique for two reasons: 1) The probe molecules were in solution and thus reported the oxygen pressure in the immediate environment of the mitochondria. 2) Since time during which the data were taken was short (<100 ms) and the measurements could be taken twice per second, it was not necessary to add oxygen continuously to the medium to resolve the data at low oxygen concentrations. As a result, except for the first few seconds after addition of oxygen (these data were not used), there were no macroscopic oxygen concentration gradients in the sample. Thus the measured oxygen concentration was that in the medium immediately surrounding the mitochondria.

Degn and co-workers (11, 12) used a special design "respirograph" to measure the oxygen dependence of respiratory

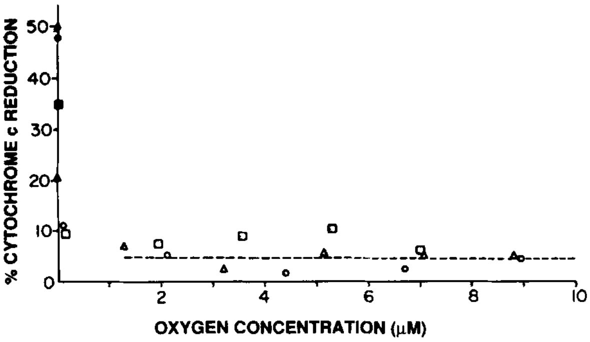

FIG. 6. **The oxygen dependence of cytochrome *c* reduction in uncoupled mitochondria.** Mitochondria were suspended as given in the legend of Fig. 4 except that 3 μM uncoupler, S-13 (5-chloro-3-*t*-butyl-2′-chloro-4′-nitrosalicylanilide), was added. The data for three cycles of oxygen depletion are given as individual points. The *dashed line* is drawn at the mean level of cytochrome *c* reduction at oxygen concentrations from 10 to 100 μM. The absorbance measurements had a response time of 2 s (0–95% response) and a delay time of less than 0.2 s. The data points were being taken each 0.5–0.8 s.

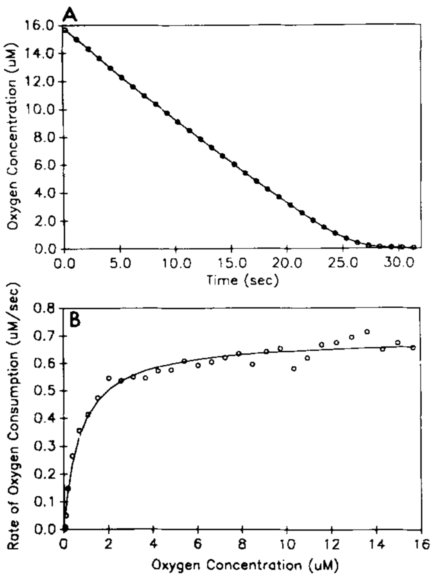

FIG. 7. **Oxygen consumption by suspensions of well coupled mitochondria at low oxygen concentrations.** Mitochondria were suspended as described in the legend of Fig. 4. *A*, the decrease in oxygen concentration at pH 7.4 and in the presence of 0.5 mM ATP. Only the experimental data for less than about 16 μM O_2 are presented. *B*, the rate of mitochondrial respiration as a function of oxygen concentration. The Asystant software was used to calculate the rate of respiration (dO_2/dt) at each point. The best fit of the data to Equation 4 (see "Results") is presented as a *solid line*. The calculated V_{max} was 0.67 μM/s and the P_{50} was 0.74 μM.

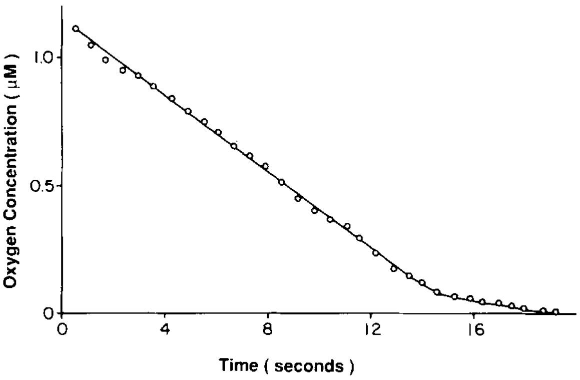

FIG. 8. **The oxygen consumption by uncoupled mitochondria at pH 7.4.** The mitochondria were suspended as described in the legend of Fig. 4 except that the protein concentration was about 0.05 mg/ml and the uncoupler S-13 (5-chloro-3-*t*-butyl-2′-chloro-4′-nitrosalicylanilide) was added to a final concentration of 3 μM. The high concentration was required because bovine serum albumin was present in the medium at a concentration of 0.4% by weight. The *straight line* drawn through the points also passes through the points from 50 to 1 μM oxygen. When the data were analyzed using the Asystant software, the apparent P_{50} for oxygen was too low to measure (less than 0.09 μM).

rate and cytochrome reduction in suspensions of isolated mitochondria. This involved allowing the mitochondrial suspension to become anaerobic and then slowly raising the oxygen concentration (measured with an oxygen electrode). Increasing the oxygen concentration was accomplished by increasing the oxygen pressure in the gas phase over the stirred suspension. Twenty to sixty minutes were used to raise the oxygen concentration in the medium from 0 to approximately 20 μM. They reported that the P_{50} for oxygen was 0.5 μM for coupled mitochondria and decreased to less than 0.05 μM when the mitochondria were uncoupled. Their measurements were restricted to oxygen concentrations less than 20 μM and to neutral pH. The experimental data were quite similar to those presented in this paper for pH 7.0 and low oxygen concentrations. Chance and co-workers (13, 15) used the luminescence of *Photobacterium phosphoreum* as an indicator of oxygen concentration and a reoxygenation method similar to that of Degn and Wohlrab (11) to study the oxygen dependence of mitochondrial oxidative phosphorylation. These authors reported that the P_{50} values for cytochrome *c* oxidation and for respiration were approximately 0.04 μM for coupled (high "energy state") and 0.5 μM for uncoupled mitochondria with succinate as substrate (14). The P_{50} values increased linearly with increasing respiratory rate (14, 15). The data of Chance and co-workers (13–15) are inconsistent with the measurements by both our laboratory and that of Degn and co-workers (11, 12), and the reason remains to be established. A critical control, measurements of the P_{50} at different concentrations of mitochondria in the same metabolic state, was not presented. This leaves open the possibility that the increase in P_{50} was due to increased rates of oxygen consumption *per se* and not to the change in metabolic state of the mitochondria. In the present experiments the P_{50} for coupled mitochondria was independent of the concentration of mitochondria. The P_{50} for uncoupled mitochondria was lower than that of coupled mitochondria even when the respiratory rate of the latter exceeded that of the former.

There have been other measurements of the oxygen dependence of respiration by suspensions of isolated mitochondria and cells in which mitochondrial oxidative phosphorylation was primarily responsible for respiration (see, for examples, 8–10, 17–21). Measurements of suspensions of isolated mitochondria in the presence of high levels of glucose and hexokinase (maximal rates of ATP synthesis) were reported to give P_{50} values of 0.04–0.08 μM (16). This value would be expected to approximate that of uncoupled mitochondria because the ATP concentrations are very low, and both ADP and inorganic phosphate are high.

Limitations Inherent to the "Steady State" Approaches to Measuring the Oxygen Dependence of Mitochondrial Function—The primary weakness of the steady state approaches described relates to the techniques used to overcome the slow response times of the measurement systems. In essence, it is necessary to design an experimental system in which the changes in oxygen concentration were slow enough to be measured. In each case this was done by placing the mitochondrial or cellular preparation in a volume of liquid covered by a gas phase and oxygenating the aqueous suspension by stirring with a mechanical paddle or bar. The oxygen dissolved in the surface of the liquid and then was stirred into the rest of the solution. The chamber was a square or multisided optical cuvette, and the oxygen measurements were made at (1, 16) or near (13–15) the bottom of a sample at least 2 cm deep. Because of this geometry, an added reagent requires at least 1–2 s for mixing even when it is forcefully injected directly into the region of maximal stirring. In the steady state experiments the oxygen was layered onto the liquid surface, and in this case several seconds are required to obtain equilibrium distribution. The mixing time can be visualized by adding a colored solution and observing the well defined concentration gradients near the surface and in the vortex of the stirrer. Continuous addition of oxygen would not be expected to generate a true steady state in the systems it was used to study. The oxygen consumption rates were greater than 0.25 μM/s, often near 1 μM/s, and substantial concentration gradients would be present in the stirrer vortices along which the oxygen traveled down into the lower portion of the sample. Thus, parts of the sample would be exposed to oxygen concentrations higher than would the oxygen-measuring device. Vigorous stirring helps to minimize the oxygen gradients but cannot fully eliminate them. Unfortunately, in each case the mixing characteristics were peculiar to the original apparatus and cannot be experimentally tested.

Factors Affecting the Oxygen Dependence of Mitochondrial Cytochrome c Oxidase—The measured dependences of the rate of mitochondrial respiration (this paper) and of the reduction of the respiratory chain components (this paper, see also Refs. 9 and 10) on oxygen concentration are not constant, but change dramatically with changes in metabolic status. Such a behavior is consistent with what is known about the mechanism of oxygen reduction by cytochrome *c* oxidase (see Refs. 9 and 10). Oxygen consumption by cytochrome *c* oxidase obeys saturation kinetics (Equation 4), but not due to formation of a simple Michaelis complex with oxygen. The internal electron transfer and energy coupling reactions act in concert to give a dependence on oxygen concentration in which the half-maximal respiratory rate (P_{50}) is a dependent variable, varying with the metabolic state of the mitochondria. In general, the P_{50} for oxygen decreases with decreasing energy state ([ATP]/[ADP][P_i]).

Physiological Importance of the Oxygen Dependence of Mitochondrial Oxidative Phosphorylation—Under physiological conditions, decrease in tissue oxygen concentration to levels which limit the ability of the cell to synthesize ATP may not affect the respiratory rate. As the oxygen concentration falls, the rate of ATP utilization by the cell tends to remain constant since the ATP consuming reactions are not dependent on oxygen concentration per se. This means that limitations in oxygen needed for the function of cytochrome *c* oxidase become evident initially in the cytoplasmic phosphorylation state ratio ([ATP]/[ADP][P_i]) and the level of reduction of its substrate, cytochrome *c*. (The level of reduction of cytochrome *c* is determined by the intramitochondrial [NAD^+]/[NADH] ratio and the [ATP]/[ADP][P_i] through near equilibrium of the first two phosphorylation sites.) Only when the changes in these two parameters are insufficient to compensate for the decrease in oxygen concentration does the respiratory rate decrease.

The fact that the oxygen dependence of mitochondrial oxidative phosphorylation extends well into the physiological range is of great importance to our understanding of many physiological functions. It allows mitochondria to function as tissue "oxygen sensors" converting information concerning the cytosolic oxygen concentration into a metabolic message ([ATP]/[ADP][P_i]) which can communicate this information to every aspect of cellular function. Indeed, strong evidence has been obtained that mitochondrial oxidative phosphorylation is a major oxygen sensor for such critical functions as regulation of blood flow (22, 23) and in regulation of respiration via the carotid body (24, 25). Since the rate of ATP utilization is not directly dependent on oxygen concentration, the result may instead be decreased cytosolic [ATP]/[ADP][P_i] and/or reduction of the mitochondrial pyridine nucleotides (for review, see Ref. 26). Changes in either of these two regulatory parameters can act to hold the rate of ATP synthesis (rate of respiration) equal to the rate of ATP utilization, albeit at a different cellular metabolic state. Indeed, failure of these regulatory changes to compensate fully for the decrease in the rate of ATP synthesis leads to catastrophic metabolic failure. This catastrophy can be avoided only if the falling tissue energy level depresses ATP utilization, establishing a new hypometabolic steady state in which nonessential ATP utilization is "turned off" but the energy level is still sufficient for essential cell functions.

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